

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

SHORT ANSWER TEST

Code of Conduct

I G. Santosh Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Santosh Kumar

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Q1. Cloud computing is the delivery of computing resources such as servers, storage, databases, networking & software over the Internet on demand, usually with pay as you use pricing.

characteristics:-

on-demand self-service

Broad network access

Resource pooling

Rapid elasticity

Explanation:-

users can create or access computing resources when ever they need.

cloud services can be accessed through the Internet using laptops

A cloud provider shares its large pool of computing resources for many customers.

Resources can be quickly increased or decreased according to demand.

Q2. cloud computing has developed gradually through several technologies changes.

Type:-

1. Hardware evolution:-

Computers initially required large physical machines over time.

Hardware becomes faster, smaller, cheaper.

2. Networking & Internet evolution:-

computer networks allowed different computers to communicate.

The Internet made it possible to access computing resources from anywhere.

3. Virtualization:-

virtualization allowed one physical server to run multiple VM.

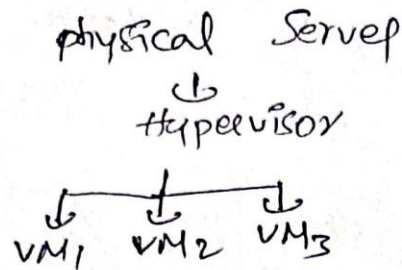
4. web & software :-

Software moved from locally installed applications towards web-based applications.

5. modern cloud platforms:-

This has enabled platforms like AWS, Microsoft Azure & Google cloud.

Q3. Server virtualization divides one physical server into multiple (VMs) using a software layer called as hypervisor.



Each VM can run its own OS.

Virtualization

Technology that creates virtual resources.

Can exist without cloud computing

Ex : - VMware

Cloud Computing

Service model that delivers computing resources.

commonly uses virtualization, but also relies on networking, automation, storage, security etc.

Ex : - AWS

- Virtualization use hardware more efficiently.
- Run multiple customers on the same physical infrastructure.
- Quickly Create & remove virtual servers.
- Isolate workloads.
- Reduce hardware costs.

Q7.

A data center is a facility containing large numbers of servers, storage systems, network equipment, power systems & cooling systems.

cloud providers operate data centers around the world.

main functions:-

1. Computing:-

- provides physical server & virtual machine.
- Runs applications & processes user requests.

2. Storage:-

- stores files, databases, backups, photos, videos etc.

3. Networking:-

- connects servers to the Internet & to other data centers.

4. Availability:-

- Multiple servers & locations help keep services available.

5. Security:-

- use firewalls, access controls, encryption, monitoring, & other security mechanisms

6. Scalability:-

- Additional Servers can be added to provide more computing capacity when demanded increases.

Q5.	meaning	IT provider	Real world ex
model			
IaaS	Infrastructure as a service	VM, storage, networking	AWS EC2
PaaS	Platform as a service	Development platform - form	Google App Engine
SaaS	Software as a service	• Complete software application	Google Docs
CaaS	everything as a service	various IT capabilities	Microsoft 365

IaaS:-

Infrastructure as a service

• provides virtual servers, storage & networking

• Ex:- AWS EC2

PaaS:- Platform as a service

• provides a platform to develop & deploy applications.

• Ex:- Google app

SaaS:- Software as a service

• provides ready to use software through the Internet.

• Ex:- Google Docs

XaaS:- Everything as a service

• provides different IT services through the Internet.

Ex:- Microsoft 365

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow